Track 1 methods report

BUB1B compound heterozygote, ranked by frozen rules and checked out of sample

Team DETA (detahub.com)

Track 1, Variant Prediction (SageBio “Rare Disease, Real Kid” MVA Hackathon 2026)

Submission date: 2026-10-20 | Version 1.3

Abstract

Row 1 of our 10-row submission for proband `PROBAND01` is BUB1B c.2210T>G p.(Leu737Ter) with c.3006T>G p.(Asn1002Lys) (GRCh38 chr15:40209701 T>G and chr15:40220612 T>G), submitted with moderate confidence. Candidates were ranked by four rule versions, each hashed (sha256) before it ran and logged in a hypothesis register of 101 append-only entries; BUB1B is rank 1 under rules v3 and v4. The frozen rules were then tested out of sample by writing public ClinVar Pathogenic/Likely pathogenic alleles into the child’s own background: 1,611 ranking trials across eight pre-registered strata, plus a second pre-registered spike-in of 779 trials that chose the row-building rule (mean rank points 80.2 against 49.1 for our draft rule). What the ranking achieves is conditional, and the condition is measurable. In recessive genes outside the primary set that match at least one of the child’s phenotype terms, rank-1 recall is 0.857 [0.814, 0.893] over 300 trials; in genes matching none it is 0.010 [0.003, 0.026] over 300. The same split holds inside the primary stratum, as a descriptive stratification that was not the pre-registered test: 160 of 163 trials (0.982) in phenotype-matched genes inside the ranking universe, and 0 of 121 in genes matching no term, every one of which landed at rank 2. The pre-registered test was the mixture of those two populations and its bar was not met: rank-1 recall 0.533 [0.477, 0.589] across the 14 of 18 MVA and phenocopy genes that held an eligible allele, against a pre-registered bar of 0.80, with a gene-clustered interval of [0.27, 0.80] and top-10 recall 0.947. Rank-1 recall for other BUB1B pairs is 1.000 over 100 trials. Four limits qualify all of this. Rank 1 depends on the ClinVar evidence tier: with ClinVar hidden no novel causal reaches rank 1 (0.000), although 0.790 reach the top 10. One MVA gene (RNU4ATAC) carries the highest phenotype score in the set and is never ranked at all, because it is non-coding and invisible to an exonic pipeline. Phase is not measured from short reads, so the compound-heterozygous configuration is a hypothesis. And the three calibrated variant-effect models disagree on p.Asn1002Lys. The EPCR values are calibrated from the spike-in.

Proband `PROBAND01`. Coordinates are GRCh38. The Markdown source of this report is in the team’s GitHub repository.

1. Answer and confidence

Submitted at row 1: **BUB1B c.2210T>G p.(Leu737Ter), chr15:40209701 T>G, with c.3006T>G p.(Asn1002Lys), chr15:40220612 T>G** (canonical transcript `ENST00000287598`, GENCODE v44).

Confidence: moderate.

- Evidence for: both calls are heterozygous in our alignment (47 reads, 55% alternate; 27 reads, 44% alternate; mapping quality 60; both strands). Both are rare in gnomAD v4.1 exomes (AF 7.9e-5 and 6.8e-7). The nonsense allele is Pathogenic/Likely pathogenic in ClinVar (Variation ID 533901, “criteria provided, multiple submitters, no conflicts”, listed conditions include mosaic variegated aneuploidy syndrome 1; ClinVar VCF fileDate 2026-10-11). BUB1B is rank 1 under our frozen rules v3 and v4 and

matches 3 of the child's 8 HPO terms (small for gestational age, rhabdomyosarcoma, short stature). Truncating plus missense BUB1B alleles are the published MVA pattern (Hanks 2004, PMID 15475955; Suijkerbuijk 2010, PMID 20516114), and a mouse model of the genotype class shows that BubR1 allelic effects drive phenotypic heterogeneity (Sieben 2020, PMID 31738183).

- Not established: phase. That the two alleles are in trans is inferred, not measured (section 5). The child's own nucleotide change has no ClinVar record, but the identical protein change does: c.3006T>A reaches the same p.Asn1002Lys substitution and is classified Uncertain significance, criteria provided by a single submitter (VCV004600147, last evaluated 2025-09-19, and present in the ClinVar release used here). The only other allele recorded at that position, c.3006T>C, is synonymous and Likely benign. There are no functional data on the child's allele, and the three calibrated predictors disagree on it (section 7).

The submitted answer is the best-supported candidate on this evidence; it is not a demonstrated cause.

2. Submitted rows (restated, because the Space does not store the CSV)

The CSV has 10 rows, each with proband_id `PROBAND01` and finding_type "primary". The EPCR values come from the spike-in calibration in section 6.

row	gene	rule tier	variants	consequence	EPCR	notes column
1	BUB1B	A	chr15:40209701 T>G + chr15:40220612 T>G	nonsense + missense	0.533223	BUB1B; rule v4 rank 1; tier A; nonsense + missense pair; phase not measured
2	LZTR1	A	chr22:20996720 C>G	nonsense	0.413611	LZTR1; rule v4 rank 2; tier A; nonsense
3	GNRHR	A	chr4:67753920 C>T	missense	0.001641	GNRHR; rule v4 rank 3; tier A; missense
4	POU6F2	B	chr7:39339680 C>CA + chr7:39339682 C>CAG	frameshift + frameshift	0.001631	POU6F2; rule v4 rank 4; tier B; frameshift + frameshift pair (ROW-MAP P5)
5	EIF4EBP1	B	chr8:38057205 C>CA + chr8:38057230 C>A	frameshift + missense	0.001621	EIF4EBP1; rule v4 rank 5; tier B; frameshift + missense pair (ROW-MAP P5)
6	HRNR	B	chr1:152214827 A>C + chr1:152214830 C> CATGTTGGCCACAG	missense + frameshift	0.001611	HRNR; rule v4 rank 6; tier B; missense + frameshift pair
7	IGBP1	C	chrX:70134785 A>C	missense	0.001601	IGBP1; rule v4 rank 7; tier C; missense
8	ADAMTS1	C	chr21:26844275 GCCCCTTC>G + chr21: 26844283 GTCCTC GCCCTCAGT>G	two frameshift deletions	0.001591	ADAMTS1; rule v4 rank 8; tier C; two frameshift deletions
9	AGAP3	C	chr7:151086835 T>TGGGG	frameshift insertion	0.001581	AGAP3; rule v4 rank 9; tier C; frameshift insertion
10	FOXO6	C	chr1:41382200 G>(G plus 200 inserted bases)	frameshift insertion	0.001571	FOXO6; rule v4 rank 10; tier C; frameshift insertion

The CSV as submitted, `DETA_ranking-rule-v4.csv` (sha256 `a349d642...`):

```
proband_id,chrom_1,pos_1,ref_1,alt_1,chrom_2,pos_2,ref_2,alt_2,epcr,finding_type,notes
PROBAND01,chr15,40209701,T,G,chr15,40220612,T,G,0.533223,primary,BUB1B; rule v4 rank 1; tier A;
nonsense + missense pair; phase not measured
```

```
PROBAND01,chr22,20996720,C,G,,,,,0.413611,primary,LZTR1; rule v4 rank 2; tier A; nonsense
PROBAND01,chr4,67753920,C,T,,,,,0.001641,primary,GNRHR; rule v4 rank 3; tier A; missense
PROBAND01,chr7,39339680,C,CA,chr7,39339682,C,CAG,0.001631,primary,POU6F2; rule v4 rank 4; tier B;
frameshift + frameshift pair (ROW-MAP P5)
PROBAND01,chr8,38057205,C,CA,chr8,38057230,C,A,0.001621,primary,EIF4EBP1; rule v4 rank 5; tier B;
frameshift + missense pair (ROW-MAP P5)
PROBAND01,chr1,152214827,A,C,chr1,152214830,C,CATGTTGGCCACAG,0.001611,primary,HRNR; rule v4 rank
6; tier B; missense + frameshift pair
PROBAND01,chrX,70134785,A,C,,,,,0.001601,primary,IGBP1; rule v4 rank 7; tier C; missense
PROBAND01,chr21,26844275,GCCCCTTC,G,chr21,26844283,GTCCTCGCCCTCAGT,G,0.001591,primary,ADAMTS1;
rule v4 rank 8; tier C; two frameshift deletions
PROBAND01,chr7,151086835,T,TGGGG,,,,,0.001581,primary,AGAP3; rule v4 rank 9; tier C; frameshift
insertion
PROBAND01,chr1,41382200,G,GGCCCGCGCCGGGACGCCCGCCTACTTCGGCGGCTGCAAGGGCGGCGCCTACGGCGGGGGCGGGGGCTTCGG
GCCGCCGGCGATGGGCGCTCTGCGCCGTCTGCCCATGCAGACCATCCAGGAGAACAAGCAGGCCAGCTTCGTGCCGGCCGCGGCGCCCTTCCGCCCTG
GGGCGCTGCCCGCGCTGCTGCCGCCGCCGCC,,,,,0.001571,primary,FOXO6; rule v4 rank 10; tier C; frameshift
insertion
```

The rows were built as follows:

- Order: the top 10 genes under the frozen rule v4 (with its Amendment 1), applied to the corrected consequence table and not changed by judgement.
- Content: a gene with one candidate call gives a single-variant row. Otherwise the pair follows the row-building rule chosen by a pre-registered test (`PRE-REG-ROWMAP`, section 6). The anchor is a candidate call with a ClinVar Pathogenic/Likely pathogenic assertion (truncating first, then higher per-variant score, then lower position); a homozygous anchor is a single row; otherwise the partner is the best other heterozygous candidate call by the same key. A gene with no P/LP call gives a homozygous call alone if one is present, otherwise its two best heterozygous calls by that key. Per-variant score: 1.0 for nonsense or frameshift, the AlphaMissense score for missense (Cheng 2023, used as a method), 0.5 otherwise. When a gene has three or more heterozygous candidate calls, its alternate pairs are listed below the lead pair; on these data no row is inserted. Compared with the draft rule used before the test, rows 4 and 5 changed: POU6F2 is now its two frameshift calls, and EIF4EBP1 keeps its frameshift call and takes the missense call with the highest AlphaMissense score instead of the lowest-position one.

  AlphaMissense pathogenicity predictions (hg38) are Copyright 2023 DeepMind Technologies Limited, obtained from Zenodo record 10.5281/zenodo.10813168 and used under the Creative Commons Attribution 4.0 International License (CC BY 4.0, https://creativecommons.org/licenses/by/4.0/); see Cheng et al., "Accurate proteome-wide missense variant effect prediction with AlphaMissense", Science 381, eadg7492 (2023), doi:10.1126/science.adg7492. We quote a small number of unmodified score values as regression test fixtures and report statistics derived from the table; the underlying data were not modified. AlphaMissense has not been validated for, and is not approved for, any clinical use. The distributed data files still begin with a superseded CC BY-NC-SA 4.0 header line, which the licence change of 2024-03-13 did not regenerate; the record metadata and README give CC BY 4.0.
- Rows 2 to 10 are the genes the rule ranks below BUB1B; we do not claim that any of them is causal. LZTR1 and GNRHR are single heterozygous ClinVar P/LP calls (LZTR1 Variation ID 1409252; GNRHR 16030, a recessive gene). The two calls of rows 4, 6 and 8 lie 2 to 8 bp apart, which usually indicates one complex indel split across records, and those of row 5 lie 25 bp apart; FOXO6 is a likely annotation artifact. These rows carry weak evidence and were kept because the rule order is frozen.
- Two rows use calls that fail the VCF FILTER (EIF4EBP1: QD2; HRNR: MQ40 and QD2). Our candidate filter never read the FILTER column. This is disclosed here and has no effect on row 1.
- finding_type is "primary" on every row. The organizers' scorer (Space commit `60e1c30e`, `evaluation.py`) parses this column but does not use it in scoring, and the FAQ's description of

secondary rows as "set aside" does not match the code.

- The file was checked with the organizers' unmodified `evaluation.py` on synthetic answer keys only (never the real key): it parses, scorer rank equals file order, EPCR is strictly decreasing and distinct, and all 15 variants exist in the supplied VCF with exact REF and ALT. A key equal to row 1 scores 100 points and F-max 1.000; row 1's first variant with an invented partner scores 50 / 0.500; keys equal to rows 4 and 10 score 25 and 10.

3. Starting point and out-of-sample evidence

- The disease name and literature point to the MVA gene set (BUB1B, CEP57 PMID 24259107, TRIP13 and second-tier genes). BUB1B was entered into our pipeline as a named target on 9 October.
- Rule v1 was written on 17 October and rule v4 on 18 October. Rules v3 and v4 were written after the v1 and v2 results on this child were known (section 4), so their rank for BUB1B is not by itself evidence that the rules generalise.
- The out-of-sample evidence is the spike-in test in section 6: the same frozen rules rank known causal variants written into this child's background, and the test reports where they fail.

4. Methods

Design. Every test was written into a hypothesis register, with a falsifier, before it ran, and its outcome was recorded as ruled out, completed without a result, or not testable. Results were appended, never edited. Ranking, calibration and spike-in rules were hashed (sha256) before they ran; revisions are dated with their reasons.

Data. Organizer dataset SageBio/mva-hackathon-2026-data at commit `59e322d` (a later commit changed only the README): Sentieon GRCh38 VCF (5,012,204 records; sha256 `bec64c68...42175`, equal to the Hugging Face stored hash), clinical document, FASTQ. We aligned the reads ourselves (PROBAND01.bam) for read-level checks.

Exonic frequency pass. All 87,198 non-reference calls in protein-coding exons (GENCODE v44 basic, +/-3 bp) were classified by gnomAD v4.1 exome frequency from regional reads of the public files (no child coordinates sent): common or low-frequency 71,416, rare 839, unobserved 14,943. Consequence was computed from the reference on 19,690 canonical transcripts. After a bug fix (section 8), 481 distinct protein-altering calls are rare or unobserved on the any-transcript definition (485 call-transcript rows, 3 of them chrM and excluded); the same table gives 478 taking the first canonical transcript per call, which is the definition section 8 uses for the withdrawn earlier figure.

Paralog-family exclusion, declared in the frozen rule before the run and carried unchanged through v2 to v4: gene symbols beginning HLA-, TAS2R, MUC, LILRA, LILRB, OR followed by a digit, ZNF, PRDM9, NBPF or SERPINA are reported separately and are not ranked. Under v1 the exclusion removed 189 candidate calls in 39 genes and left 56 genes ranked; once the v2 male-X and cluster filters are added, 277 candidate calls reach the ranking. The genome-wide compound-heterozygous pattern (a rare loss-of-function-type call and a rare missense call in the same gene, phase unknown) does not single out BUB1B: 13 genes carry the pattern on the pre-fix table and the same 13 carry it on the corrected table, so 12 genes besides BUB1B show it, most of them in paralogous or highly variable families.

Ranking rules, in the order written:

- v1: weighted consequence score. BUB1B rank 8; a shuffled-label baseline put it at 7. Flaws recorded: chrX treated as diploid in a male child; an artifact cluster.
- v2: male-X and cluster filters. BUB1B rank 6; shuffled baseline 4. Data-only weighting does not separate BUB1B from background.
- v3: evidence tiers. A = a call classified Pathogenic/Likely pathogenic in ClinVar; B = loss-of-function-type plus missense calls in one gene; C = other genes passing v2. BUB1B rank 1. Without the ClinVar

tier it is rank 6, and with ClinVar labels shuffled it leaves tier A. Rank 1 therefore rests on existing knowledge.
- v4: within each tier, order by the number of the child's 8 HPO terms matched to the gene's HPO annotations through is_a ancestry (HPO release 2026-09-01). Amendment 1 (dated, before the corrected run) excludes chrM calls. BUB1B rank 1; only 3 of 52 candidate genes match any term. The recorded v4 hash covers the rule text as first frozen; Amendment 1 was appended afterwards, dated and before the corrected run, and the amended file's hash is recorded beside it.
- The 8 HPO IDs in the rule (`HP:0000121`, `HP:0001508`, `HP:0001518`, `HP:0001622`, `HP:0002859`, `HP:0003202`, `HP:0004322`, `HP:0200067`) were checked against the organizers' clinical document: exact match, labels equal to hp.obo names.
- Script defects found and handled without changing any rule: the rerun script for v4 on the corrected table had dropped the Amendment 1 chrM line (three MT- genes appeared at ranks 7, 9, 10); the submission applies the amendment as written, which moves no other row. The frozen v4b script reads the pre-fix v3 output; applied to the fixed v3 output it gives the same top 10.

Other alternatives tested on the child's data (all ruled out at the level tested): CEP57 c.915_925dup11 missed by the caller; the BUB1B upstream allele `rs576524605` (reference at 45 reads; the allele class is the one reported in PMID 24344301); a BUB1B intronic mobile element; structural variants at the BUB1B locus (Manta 1.6.0, Chen 2016, PMID 26647377; zero calls within BUB1B +/-50 kb, filtered or unfiltered); low-VAF mosaic signal; autozygosity; TRIP13 and CEP57 biallelic loss; second-tier MVA genes (SMC5 PMID 36333305 and PMID 41374403, SLF2 PMID 36333305, MAD1L1 PMID 36322655, CENATAC PMID 34009673, TRIM37 PMID 41702694, MAD2L1BP PMID 37796616, CEP192 PMID 37981762, CDC20 PMID 33094908; only common or single calls); hemizygous NAA10 loss; arm-level and whole-chromosome copy change at our detection limit (about 14% clonal fraction); homozygous SMN1 loss.

The genome-wide dosage test is stated separately, because its two outcomes are different. Under the structural-variant rule, 113 dosage-sensitive genes carry a passing call, a count that noise across large genes is expected to produce, and none was carried to validation or phenotype matching: that test is **NOT CONCLUDED**, not negative. The two large DICER1-region calls were tested directly and ruled out by depth (1.02 and 1.03 times the autosome median, against about 0.5 and about 1.5 for a real heterozygous deletion and duplication), and that is the only part of this line that is ruled out.

Pre-registered checks added on 20 October:

- P2, a sweep of 18 MVA and phenocopy genes (including RNU4ATAC, RNU6ATAC, ZRSR2, SNRNP48, ESCO2, PCNT, CENPF): only BUB1B was flagged. Gaps: non-exonic calls not interpreted; ZRSR2 chrX calls need a hemizygosity check; depth over RNU4ATAC and RNU6ATAC not checked; local gnomAD sites files were missing, so frequencies came from our exonic table (disclosed deviation).
- P3, the two published CENATAC family alleles (de Wolf 2021, PMID 34009673): both absent (51 reference reads at c.803; no deletion reads at c.805+2_805+19del18).
- `PRE-REG-D`, placement of N1002K next to the characterised BUB1B missense alleles I909T and L1012P (Suijkerbuijk 2010): the pre-registered falsifier was met (section 7).
- `PRE-REG-ROWMAP`, the rule that builds a gene's row from its candidate calls: rule P5 with construction C3 adopted (section 6).

5. Phase (P1): unresolved

Read-backed phasing was pre-registered with a fixed rule (a chain of heterozygous sites, at least 5 bridging read pairs per link, at least 90% concordance). One heterozygous SNV lies between the two calls, leaving gaps of at least 6,769 bp and 4,142 bp. Of 2,029 read-1 templates in the window, the largest insert is 1,590 bp, so no fragment spans either gap and no chain can be built: **UNRESOLVED**. The limit is the short-read fragment length.

Deviation: the script that produced this result checks fragment span against gap length only; it does not

count allele-carrying read pairs at base quality 20 and has no proper-pair filter, unlike the pre-registered text. The result does not change, because zero fragments span either gap under any allele-level filter.

Resolving phase needs long reads spanning about 11 kb or parental genotypes; the dataset has neither. Row 1 is therefore a compound-heterozygous hypothesis. The scorer treats a row as an unordered variant set and does not use phase.

6. Out-of-sample test: spike-in of known causal events

Question: does the frozen ranking put known causal events at rank 1, or did it work only for this child? The test was pre-registered and hashed before any spiked ranking ran (20 October), then audited independently (83 of 83 rebuilt trials matched).

Design: the child's two BUB1B rows were removed; public ClinVar Pathogenic/Likely pathogenic alleles were written into the child's own background as biallelic genotypes; the frozen consequence step and rules v3 and v4 were rerun per trial. This tests ranking only, not read-level calling or phasing, and the child's eight phenotype terms are held fixed across all 1,611 trials, so the test cannot detect overfitting to that phenotype vector.

stratum	trials	rank 1	top 10
A. Primary: MVA and phenocopy set, gene-uniform over the 14 of 18 genes with an eligible allele	300	**0.533 [0.477, 0.589]**	0.947
A-BUB. Other BUB1B P/LP pairs	100	1.000 [0.975, 1.000]	1.000
B. Same primary events with ClinVar hidden and no frequency (novel)	300	**0.000 [0.000, 0.008]**	0.790 [0.741, 0.833]
E. Secondary: recessive genes outside the MVA set with at least one phenotype match	300	0.857 [0.814, 0.893]	0.987
D. Control: recessive genes with no phenotype match	300	0.010 [0.003, 0.026]	0.957
C1, C2-HOM and C2-CH. Stop-loss alleles (ClinVar and synthetic)	311	labelled stop_lost in 0 of 311	

(Jeffreys 95% intervals. The eight pre-registered strata appear here as six rows: the stop-loss row pools the three pre-registered stop-loss strata, C1, C2-HOM and C2-CH.)

Findings:

- **Negative robustness result.** The pre-registered bar was rank-1 recall of at least 0.80; we measured 0.533, so the bar was not met, and the hypothesis register records that outcome under its own label, FALSIFIED. (The same pre-registration predicted 0.4 to 0.6 before the run.) Four of the 18 genes in the set (RNU6ATAC, SNRNP48, CEP192, MAD2L1BP) held no eligible public Pathogenic/Likely pathogenic allele to write in and contributed no trial, so the estimate is over the 14 genes that could be sampled. Five MVA genes (CDC20, CENATAC, CENPF, SLF2, SMC5) never reach rank 1 because HPO does not annotate them with any of the child's terms; the non-coding RNU4ATAC is invisible to the exonic pipeline.
- **Gene clustering.** Outcomes are nearly fixed per gene. Resampling genes rather than trials gives 0.533 with a 95% interval of [0.27, 0.80]; the trial-level intervals hold only for this gene mix.
- **ClinVar dependence.** With ClinVar hidden, no novel causal reaches rank 1 (0 of 300), although 79% reach the top 10. Rank 1 in this pipeline needs the ClinVar tier.
- **Control blocking.** Unrelated recessive genes reach rank 1 in 3 of 300 trials, all through ClinVar records shared with an overlapping phenotype-matched gene (0 of 292 otherwise). Without the phenotype layer (rule v3) they reach rank 1 in 76%. The phenotype layer gains this specificity at the cost of MVA-set recall (v4 0.533 vs v3 0.840).

- **BUB1B specifically.** Other BUB1B P/LP pairs rank 1 in 100 of 100 trials, so the child's rank 1 does not depend on these exact alleles.
- **Stop-loss blind spot.** The consequence classifier reads only GENCODE exon and CDS features and ignores the stop_codon feature, so a stop-codon change is labelled UTR and is never scored as stop-loss (0 of 311). A stop-loss causal variant would be missed, or would receive at most half credit through a partner.

What determines rank 1 in this design, stated descriptively. Two pre-registered strata point at the phenotype layer: recessive genes outside the MVA set that match at least one of the child's HPO terms reach rank 1 in 0.857 [0.814, 0.893] of 300 trials, and genes that match none reach it in 0.010 [0.003, 0.026] of 300. Those two strata are different gene sets, so they differ in gene membership as well as in phenotype match, and with outcomes nearly fixed per gene that contrast is suggestive, not a controlled single-variable result. The controlled comparison is the v3 against v4 ablation, which changes one rule inside the same genes and the same trials: the phenotype layer costs recall in the primary stratum, 0.840 under v3 against 0.533 under v4, and buys specificity in the control stratum, where unrelated genes reach rank 1 in 76% of trials under v3 against 1% under v4. The same split runs through the primary stratum. Taking the per-gene outcomes and grouping them by phenotype score, the eight genes with a score of at least 1 that are inside the ranking universe give 160 rank-1 hits in 163 trials (0.982), and the five genes with a score of 0 give 0 in 121, every one of them landing at rank 2 behind LZTR1. RNU4ATAC is counted in neither group and is reported on its own: it carries the highest phenotype score of any gene in the set, 4, and was never ranked in any of its 16 trials, because it is non-coding and an exonic pipeline cannot see it. All 160 rank-1 hits in the primary stratum come from the phenotype-matched group, so the pre-registered 0.533 is the mixture of those two populations and nothing else. The grouping reconstructs the stratum's published distribution exactly, which is the check that it is a real division and not a convenient slice: 163 + 121 + 16 = 300 trials, and 160 rank-1, 124 rank-2 and 16 never ranked = 300 outcomes, with the 124 at rank 2 resolving as the 121 phenotype-score-zero trials plus three TRIM37 misses. **This grouping is descriptive and was not the pre-registered test. It does not rescue the bar, which was not met on the stratum the pre-registration named.** What it establishes is the condition under which the rule reaches rank 1, and that the condition is measurable before the rule is run: it is whether HPO annotates the gene with a term the child has.

EPCR. Row r gets the smoothed fraction of primary-stratum trials whose causal landed at rank r, (k + 0.5)/301, made monotone, with a 1e-5 step per row to keep values distinct (frozen rule). Row 1: 160/300 gives 0.533223. Row 2: 124/300 gives 0.413611. Rows 3 to 10: 0/300 each, giving 0.001641 down to 0.001571.

- EPCR is a calibration of the ranking, not a posterior for this child. For BUB1B events the measured rank-1 rate is 1.000, so 0.533 is conservative for row 1.
- **EPCR at row 2 measures "the slot behind the top phenotype-matched gene", not LZTR1's own probability of being causal.** In the masked background the rank-2 mass is MVA genes with no HPO match sitting behind LZTR1.
- The scorer sorts by EPCR and sweeps EPCR thresholds; with strictly decreasing distinct values only the order matters.

Row-building rule (`PRE-REG-ROWMAP`). The draft rule used before this test paired, in any gene with more than two candidate calls, the highest-weight loss-of-function-type call with the lowest-position missense call. A second spike-in, pre-registered and hashed before it ran (sha256 `9f7a440d...`, frozen 2026-10-20 11:31:36Z; first trial ranking 11:37:14Z), compared that draft rule with five mechanical alternatives under three row constructions. Each trial wrote a known compound-heterozygous causal pair and one or two decoy heterozygous calls (public ClinVar alleles not classified P/LP) into the causal gene in the child's background, and every trial file was scored with the organizers' unmodified `evaluation.py`.

primary strata (779 trials, 201 genes)	adopted rule (P5, construction C3)	draft rule (construction C1)
mean rank points	80.2 [73.3, 86.8]	49.1 [44.8, 53.4]
exact pair on row 1	0.626 [0.592, 0.660]	0.095 [0.076, 0.117]
F-max	0.897	0.525

(Gene-clustered bootstrap 95% intervals for rank points; Jeffreys 95% intervals for the row-1 rate.)

- The difference in mean rank points is +31.1 (95% stratified gene-cluster bootstrap interval [26.5, 35.5]; the pre-registered Bonferroni gate interval for 17 comparisons, [24.3, 37.7], excludes zero), so the pre-registered decision rule adopted P5 with construction C3. One deviation, in a sanity-check implementation, affected no decision.
- Near-tie (post hoc): the adopted rule beats two other mechanical rules (the top two calls by score, P2, and the weaker-allele rule, P3) by only +0.13 points [0.03, 0.26], and a fixed filter rule (P1) by +1.32 [-1.36, 3.77]. Almost all of the gain comes from dropping our draft rule, not from the choice among its replacements.
- Why the draft rule failed (post hoc diagnostic): it always paired one loss-of-function-type call with a missense call, so a missense decoy took the place of a second truncating or splice-region causal allele. In the 444 trials with such a causal pair it built the correct pair in none (0 of 444).
- Limits: the decoys are public ClinVar alleles, almost all missense, not the child's own passenger calls, and the ordering among the top rules is within noise.
- Real data: the BUB1B interval holds 14 calls. Only the two submitted calls lie in a protein-coding exon and every other call is at least 905 bp from one, so each rule sees exactly two BUB1B candidate calls and row 1 is identical under all 18 rule-by-construction combinations. The adopted rule changes rows 4 and 5 only (section 2); with the BUB1B pair as the key, the automated score is unchanged.

7. Variant-effect models

A calibration rule was written and hashed before any model ran: ClinVar controls chosen by a fixed hash rule without model scores (BUB1B excluded), AUC with a 2,000-resample bootstrap 95% CI; "informative" needs a lower bound above 0.7, 0.6 to 0.7 is exploratory, below 0.6 is not used. Child percentiles are reported only for informative models.

model	AUC (95% CI)	verdict	N1002K percentile among controls
ESM-1v (Meier 2021, bioRxiv, doi:10.1101/2021.07.09.450648), 98 missense controls	0.964 (0.916-0.998)	informative	8.2% (benign-leaning)
ESM-2 650M (Lin 2023, PMID 36927031), 98 missense controls	0.875 (0.805-0.935)	informative	61.2%
AlphaMissense, public table (Cheng 2023, PMID 37733863), 96 matched	0.921 (0.861-0.969)	informative	76.0%
Evo 2 7B (Brixi 2026, PMID 41781614), 198 controls	0.677 (0.602-0.748)	exploratory only	not reported

- AlphaMissense (Cheng 2023) predictions are used under CC BY 4.0 (licence and attribution in section 2); the AUC and percentiles in this section are statistics derived from the table, which was not modified.
- The informative models disagree at N1002K, so the computational evidence does not settle its effect. AlphaMissense may be optimistic on ClinVar controls.
- Protein models used missense controls only (they cannot score a stop): a deviation from the pooled design.

- `PRE-REG-D`: I909T and L1012P are poorly expressed BUB1B substitution mutants whose function was restored by overexpression (Suijkerbuijk 2010, PMID 20516114). Percentiles (ESM-1v / ESM-2 / AlphaMissense): I909T 60.2 / 60.2 / 54.2; L1012P 39.8 / 66.3 / 72.9; N1002K 8.2 / 61.2 / 76.0. Two models place N1002K outside the range of the two reference alleles, so the pre-registered falsifier of the placement hypothesis was met.
- Evo 2 deviations (disclosed): the checkpoint needs a Transformer Engine pickle override (`NVTE_ALLOW_UNSAFE_PICKLE_EXTRA_STATE=1`), set for one process after the weights' sha256 matched the Hugging Face object id; FP8 input projections were turned off because the RTX 3090 cannot run FP8, so inference ran in bf16, which differs numerically from the trained path; one GPU only. Raw deltas for the child (nonsense -46.78, missense -11.68) are uninterpreted output.
- A regulatory model (Borzoi; Linder 2025, PMID 39779956) scored at chance on its controls (AUC 0.496) and is not used.

Mechanism of the nonsense allele (computed from the transcript, not tested in the child): the stop lies in exon 17 of 23, about 747 nt upstream of the last exon-exon junction. A premature stop more than 50 to 55 nt upstream of that junction is expected to trigger nonsense-mediated decay (Nagy and Maquat 1998, PMID 9644970; figure as stated in PMID 16326120), so the allele is predicted null. TGA readthrough is drug- and context-dependent (PMID 39174735).

8. Corrections and limitations

- Consequence bug: for minus-strand transcripts the classifier reversed the coding segments before reverse-complementing, changing 5,101 of 74,741 rows (BUB1B is plus-strand and unaffected). It was fixed in a separate script and all rule versions were rerun; rank 1 is unchanged in all four. The fix was validated on public data only: agreement with ClinVar's own consequence terms on 6,000 SNVs rose from 0.589 to 0.994 on minus-strand transcripts and stayed 0.998 on plus-strand. Counts of protein-altering rare or unobserved calls depend on the counting definition, so both are given: on the corrected table, 478 taking the first canonical transcript per call and 481 taking any transcript. The earlier figure of 487 is the first-transcript count on the pre-fix table, where it reproduces exactly; it is withdrawn as a figure for the corrected table, not as a miscount.
- Limits: phase not measured; no parental data; stop-loss blind spot; indels classified by anchor position; canonical transcripts only; splice region limited to intronic +/-2; gnomAD exomes only; FILTER column not used; structural calls from one short-read caller; full ACMG secondary-findings screen not run; no positive control exists in this call set.
- Secondary or incidental findings: none was submitted as secondary. LZTR1 (row 2) is a heterozygous nonsense call classified Pathogenic/Likely pathogenic in ClinVar, with listed conditions that include Noonan syndrome and LZTR1-related schwannomatosis; GNRHR (row 3) is a heterozygous pathogenic call in a recessive gene. Both are reported only because the frozen rule ranks them; we make no clinical claim about them.

9. Data, tools and AI use

Public data: gnomAD v4.1 exomes (regional reads of public files) and the gnomAD v4 API for 22 family-gene variants; ClinVar GRCh38 VCF (fileDate 2026-10-11); GENCODE v44 basic and v50; HPO (release 2026-09-01); ClinGen dosage list (16 October 2026); Ensembl REST; dbSNP; PubMed; AlphaMissense public scores (hg38, Zenodo record 10.5281/zenodo.10813168, CC BY 4.0; attribution in section 2); public model weights for ESM-1v, ESM-2 and Evo 2. No proprietary data.

Data handling: analysis ran on a local machine (2x RTX 3090). The only step that sent child variant coordinates off that machine was the 22-variant gnomAD API lookup.

Compute, run time and cost (times measured from tool logs and file timestamps; energy and rental figures are estimates). The workstation has an AMD Ryzen 7 9800X3D (16 threads), 48 GB RAM and 2x NVIDIA RTX 3090 24 GB, used one GPU per job. No cloud or rented compute was used. FASTQ-to-BAM alignment

with Parabricks 4.7.1 fq2bam took 82 min on one GPU; Manta SV calling took 26 min (12 threads, 16 GB); the genome-wide exonic gnomAD v4.1 frequency lookup took about 3.5 h (network-bound, 4 workers). Consequence annotation, the four frozen ranking rules, read-level checks and calibration statistics took under 10 min in total. What the harness would run is a ladder of requirements, as its own planner reports on a clean tree at commit `ab2de8d28ac2`, where it declares 53 steps, 6 of them procedures carried out outside the harness: with the organizers' dataset, the public references and nothing of ours supplied, 16 of the 53 in the quick tier and 18 in the full tier; allowing network access raises the full tier to 46, because the 3.5 h gnomAD lookup then rebuilds the frequency table the exonic and ranking chain needs; supplying our stored child-derived intermediates instead (exonic calls, gnomAD frequency parts, corrected consequence table, two ranking summaries) gives 38 quick and 44 full; adding the held-back CNV rule file and our stored model-score tables (Evo 2 controls, protein scores) gives 41 quick, which is every quick-tier step, and 45 full, or 47 with network access, which is every step the harness itself executes. The quick tier is CPU only; the full-tier figures include 4 GPU steps, the largest 16 GB VRAM, on public model weights. Those are planner figures, not runs: 37 of the 41 quick-tier steps carry a wall time measured on our machine on 2026-10-20 and those sum to about 10 min, and the tier was not re-timed at this commit. The measured outsider figure is separate: a clean-room replication executed 14 steps in about 6 min with nothing supplied beyond the repository, the organizers' data and the public references. ESM-1v and ESM-2 scoring took about 3 min, Evo 2 7B scoring of 198 calibration variants about 4 min and Borzoi about 7 min, each on one GPU. The whole pipeline takes about 6 h serial (about 4 h with the frequency lookup running alongside alignment) and 1.7 GPU-hours. The ranking alone (VCF in, ranked list out) needs no GPU and under 2 min of local compute plus the frequency lookup. There was no compute bill. Electricity for the pipeline was about 0.8 kWh (about 25 JPY, USD 0.17, at 31 JPY/kWh; GPU power sampled at about 200 to 270 W mean per job, 250 W assumed where not sampled); including exploratory and cross-check runs, about 4.1 GPU-hours and 2.3 kWh (about USD 0.47). Hardware cost is not amortised. On a rented GPU at USD 2.19/h the full pipeline would cost about USD 8 to 12. AI assistance ran on a flat-rate Claude Max subscription: about 35 h of active agent time over 14 days and about 2.0 M output tokens, with no per-token charge.

Generative AI (provider, plan, data setting and use, as confirmed by the team on 2026-10-20; not independently verified):

- Anthropic, Claude Code: Claude Max, consumer plan; model training off ("Help improve Claude" off); no ratings or feedback given. Model ids recorded in the session transcripts: `claude-opus-5-5`, `claude-sonnet-5-5`, `claude-haiku-5-5` and `claude-opus-5`. Used to write and run the analysis scripts and to draft text; these sessions processed child-derived variant data (coordinates and genotypes).
- OpenAI, ChatGPT consumer app with Codex: training off; no ratings. Model ids recorded: `gpt-6-luna` (session of 2026-10-10) and `gpt-6.1-sol` (two review runs on 2026-10-18). Used for review and research.
- Google, Gemini 3.1 Pro Deep Research: business account; data use off; no ratings. Used for literature research.
- Perplexity, xAI Grok and DeepSeek V4: used for general background research and outside critique; no project data was shared with them; plan and data-handling settings were not recorded.
- No content was flagged for safety review.

No output of a chat or agent AI assistant was used as evidence. Claude Code wrote and ran the analysis scripts under human direction, and every analysis number in this report comes from a script. Most of those scripts are in the repository and run under its replication harness. The exceptions: the alignment, Manta, SMN and Borzoi steps are external to the harness; the Manta launcher (it contains our local folders and the read-group sample label) and the SMN scripts as run (they still carry local path constants) are held back, while the Borzoi scripts are included as run but are not run by the harness. The three tests behind this report's out-of-sample evidence now stand in three different places. The spike-in test with its EPCR calibration (section 6) is covered by harness steps that rerun its 1,611 trials and check its strata, its EPCR vector and its seven pre-registered decisions against the recorded values, with the trial tables

compared by sha256. The row-building test (`PRE-REG-ROWMAP`) is a declared step that the harness does not run, because it needs the organizers' scorer and two of its own gates compare against files we hold back. The scorer checks on the submitted file are not covered by the harness at all, and need code that is not among the published scripts. The run-time and energy figures come from tool logs and file timestamps. ESM-1v, ESM-2 and Evo 2 are protein and genomic language models whose outputs are variant scores; they ran locally, and AlphaMissense scores came from its public table.

Reproducibility: the repository contains the analysis code, the replication harness with the expected value of every automated step, the frozen rule files with their sha256 (`RANKING_RULE_v1` to v4, `SV_RULE_v1` and `CALIBRATION_RULE`; `CNV_RULE_v1` as its sha256 only, because its frozen text names a local file), the spike-in pre-registration, a public copy of the ROW-MAP pre-registration (the frozen file has sha256 `9f7a440d...`; the pre-registration log records that hash and the hash of the copy), a redacted pre-registration log with self-recorded times, tests on synthetic data and the environment files. The hypothesis register is not in the repository, because it holds read-level values and the full working record rather than only the submitted content. Submitted CSV sha256 `a349d642fb4597ea2ebe1b032cf40e3ea2b962e80860e7cc9538164944d78e4f`. Input checksums: VCF `bec64c68d3ad10d05b8071d07cdf87d217013d66d3ba02e24b505035c9d42175`; VCF index `6f8fed62f11c475fc63a8e2b50925ffc7be33b6930225e821cb977951761a2e0`.

We ran that harness on our own work before submitting, and the outcome is reported here with its negative half. At harness commit `23803538190e`, where the quick tier is 34 steps of 46, the run executed all 34 and left 12 unrun because they need a GPU, the network or a container. 312 of 322 expected metric values matched. Six executed steps differed on at least one metric, 10 metric values in all, and those 10 produced errata E1 to E6; three further errata from the same run (E7 to E9) correct a manifest entry outside the register, settle the counting definition behind the withdrawn 487 figure, and state the scope the unrun steps leave unverified. E1 to E7 leave their entry's decision unchanged, E8 changes a figure's definition rather than a decision, and E9 is the scope limit, not a correction. The pre-registered decision metric of every differing step matches, so each step's recorded decision reproduces: one step differs on soft-clip counts while its RULED OUT decision matches, another on a gap length while its UNRESOLVED decision matches. **That commit and the figures given above are different harness configurations and must not be read as one**; the step count grew in later commits, and no replication report covers those. To check any of this without material of ours beyond the harness: run its quick tier at the commit named above and compare the per-metric table against the harness's own expected values, which is the same comparison the run above performed. The outsider figure of 14 steps comes from a clean-room replication, driven from the README alone against an independent copy of the repository; of the 13 steps that both runs executed, the status and every observed metric value were identical. The same walkthrough found that a literal install from the pinned requirements fails, because two pinned CUDA packages are source-only builds, and that the exonic and ranking chain cannot run at all without child-derived intermediates we do not publish.

10. What would prove the answer

- Parental genotypes showing one allele from each parent (phase and segregation).
- Long reads spanning chr15:40209701 to chr15:40220612 (about 10.9 kb).
- A functional assay of p.Asn1002Lys.

11. Methods description (organizers' template, Track 1 sheet)

The organizers' submit form accepts one report file, so the answers to their methods description template (sheet "Track 1 methods") are given here, question by question, in the template's order.

Team name

DETA (detahub.com).

Model number (fill out this form once per model/approach submitted; up to 6)

1 (the first Track 1 submission). This report describes one approach.

Please describe your model/approach in detail.

Frozen, hashed ranking rules on the organizers' GRCh38 call set, with out-of-sample spike-in tests that also set the EPCR values (sections 4 to 7).

- Exonic frequency pass: all 87,198 non-reference calls in protein-coding exons (GENCODE v44 basic, +/-3 bp) were classified by gnomAD v4.1 exome frequency and by consequence computed from the reference on 19,690 canonical transcripts; 481 distinct protein-altering calls are rare or unobserved on the any-transcript definition (478 taking the first canonical transcript per call).
- Ranking rule v4, hashed before it ran: tier A is a call classified Pathogenic/Likely pathogenic in ClinVar, tier B is loss-of-function-type plus missense calls in one gene, tier C is other genes passing v2; within each tier, genes are ordered by the number of the child's 8 HPO terms matched through is_a ancestry. The top 10 genes, in rule order, are the 10 rows.
- Row content: the row-building rule chosen by a pre-registered spike-in (`PRE-REG-ROWMAP`, rule P5 with construction C3), which anchors a pair on a ClinVar Pathogenic/Likely pathogenic call, adds the best-scoring partner and lists alternate pairs below the lead pair when a gene has three or more heterozygous candidate calls.
- Read-level checks on our own alignment, pre-registered tests of alternatives (CEP57, the BUB1B upstream allele, a mobile element, structural variants with Manta, copy number, autozygosity, SMN1, NAA10, the published CENATAC alleles and a sweep of 18 MVA and phenocopy genes) and read-backed phasing, which was unresolved.
- Spike-in: public ClinVar Pathogenic/Likely pathogenic alleles were written into the child's own background as biallelic genotypes and the frozen pipeline was rerun per trial; rank-1 recall was 0.533 [0.477, 0.589] in the primary stratum. The EPCR of row r is the smoothed fraction of primary-stratum trials whose causal landed at rank r.
- Variant-effect models under a calibration rule written before they ran: ESM-1v, ESM-2 and AlphaMissense were informative but disagree on p.Asn1002Lys; Evo 2 is exploratory only.

Result: row 1 is BUB1B chr15:40209701 T>G p.(Leu737Ter) with chr15:40220612 T>G p.(Asn1002Lys), with moderate confidence; trans is inferred, not measured.

(required) If commercially-available Generative AI was used, record the provider, the plan or tier, and the relevant setting in your methods description. A line is enough, for example, "Anthropic API, Claude Sonnet, commercial terms, no training on customer content."

Provider, plan or tier and data setting for every tool, as confirmed by the team on 2026-10-20 (not independently verified):

- Anthropic, Claude Code (`claude-opus-5-5`, `claude-sonnet-5-5`, `claude-haiku-5-5`, `claude-opus-5`): Claude Max, consumer plan; model training off ("Help improve Claude" off); no ratings given. Used to write and run the analysis scripts and to draft text; these sessions processed child-derived variant data.
- OpenAI, ChatGPT consumer app with Codex (`gpt-6-luna`, `gpt-6.1-sol`): training off; no ratings. Used for review and research.
- Google, Gemini 3.1 Pro Deep Research: business account; data use off; no ratings. Used for literature research.
- Perplexity, xAI Grok and DeepSeek V4: used for general background research and outside critique; no project data was shared with them; plan and data-handling settings were not recorded.
- No content was flagged for safety review.

Please state whether the submission file is the automated output of your computational approach (preferred), or if it has undergone downstream manual review and curation.

Automated output of frozen rules, with two documented manual steps. The order is the top 10 genes of rule v4 run by script, the row content comes from the pre-registered row-building rule run by script, and the EPCR values come from the spike-in calibration script; nothing was re-ranked by judgement. Two steps were set by hand under written rules: applying Amendment 1 (exclusion of chrM calls) as written after the rerun script was found to have dropped it, which moves no other row, and labelling every row primary (see the answer on secondary findings). Rules v3 and v4 were written after the v1 and v2 results on this child were known; the reasons are recorded in the hypothesis register.

Please describe any downstream manual review in detail.

- Rule revisions v1 to v4, each written and hashed before its run, with the reason recorded: chrX treated as diploid in a male child, an artifact cluster, data-only weighting that did not separate BUB1B from background, the phenotype layer, and chrM calls excluded by a dated amendment.
- Read-level review of the two row-1 calls (depth, allele fraction, strands, mapping quality) and a check that the 8 HPO IDs in the rule match the organizers' clinical document exactly.
- A consequence-classifier bug on minus-strand transcripts, found by review, fixed in a separate script and validated on public ClinVar consequence terms; all rule versions were rerun and rank 1 is unchanged.
- The draft row-building rule was replaced by the rule chosen in a pre-registered spike-in (`PRE-REG-ROWMAP`); this changed rows 4 and 5 only.
- Observations recorded but not acted on, so the frozen order stands: the two calls of rows 4, 6 and 8 lie 2 to 8 bp apart and those of row 5 lie 25 bp apart; FOXO6 is a likely annotation artifact; two rows use calls that fail the VCF FILTER.

Please state whether your approach only used publicly available data (preferred) or if proprietary data was also used. If proprietary data were used, where possible, please also include a submission based only on publicly available data.

Public data only. No proprietary data was used.

Please describe any public data used in detail (e.g., reference panels, population databases, literature).

- Organizer dataset SageBio/mva-hackathon-2026-data at commit `59e322d` (VCF, clinical document, FASTQ).
- gnomAD v4.1 exomes (regional reads of the public files) and the gnomAD v4 API for 22 family-gene variants.
- ClinVar GRCh38 VCF (fileDate 2026-10-11).
- GENCODE v44 basic and v50; HPO (release 2026-09-01); ClinGen dosage list (16 October 2026).
- Ensembl REST, dbSNP and PubMed (literature in the reference list).
- AlphaMissense hg38 predictions (Zenodo record 10.5281/zenodo.10813168, CC BY 4.0; full attribution in section 2 of this report) and public model weights for ESM-1v, ESM-2 and Evo 2.
- Manta 1.6.0 for structural variants.

Please describe any proprietary data used in detail.

None.

Is your approach able to output proposed pairs of compound heterozygous candidate variants, or only single candidate variants?

Pairs as well as single variants. Tier B of the ranking rule detects loss-of-function-type plus missense calls within one gene, and a row holds a pair built by the pre-registered row-building rule (a ClinVar Pathogenic/Likely pathogenic anchor and the best-scoring partner, with alternate pairs below the lead

pair when a gene has three or more heterozygous candidate calls). Row 1 is a pair. Phase is not measured from short reads (no fragment spans the gaps between the two calls in this child), so a pair is a compound-heterozygous hypothesis, not a phased call.

How did you handle secondary or incidental findings, if any? (See the finding_type column in your submission.)

Every row is marked primary; no row is submitted as secondary. The organizers' scorer (Space commit `60e1c30e`, `evaluation.py`) parses finding_type but does not use it in scoring, and the FAQ's description of secondary rows as "set aside" does not match the code, so the label was not used to separate findings. LZTR1 (row 2) is a heterozygous nonsense call classified Pathogenic/Likely pathogenic in ClinVar, with listed conditions that include Noonan syndrome and LZTR1-related schwannomatosis; GNRHR (row 3) is a heterozygous pathogenic call in a recessive gene. Both are reported only because the frozen rule ranks them; we make no clinical claim about them. A full ACMG secondary-findings screen was not run.

Please provide an estimate of run time and cost for your approach (if possible).

Run time and cost (times measured from tool logs and file timestamps; energy and rental figures are estimates). The workstation has an AMD Ryzen 7 9800X3D (16 threads), 48 GB RAM and 2x NVIDIA RTX 3090 24 GB, used one GPU per job. No cloud or rented compute was used. FASTQ-to-BAM alignment with Parabricks 4.7.1 fq2bam took 82 min on one GPU; Manta SV calling took 26 min (12 threads, 16 GB); the genome-wide exonic gnomAD v4.1 frequency lookup took about 3.5 h (network-bound, 4 workers). Consequence annotation, the four frozen ranking rules, read-level checks and calibration statistics took under 10 min in total. What the harness would run is a ladder of requirements, as its own planner reports on a clean tree at commit `ab2de8d28ac2`, where it declares 53 steps, 6 of them procedures carried out outside the harness: with the organizers' dataset, the public references and nothing of ours supplied, 16 of the 53 in the quick tier and 18 in the full tier; allowing network access raises the full tier to 46, because the 3.5 h gnomAD lookup then rebuilds the frequency table the exonic and ranking chain needs; supplying our stored child-derived intermediates instead (exonic calls, gnomAD frequency parts, corrected consequence table, two ranking summaries) gives 38 quick and 44 full; adding the held-back CNV rule file and our stored model-score tables (Evo 2 controls, protein scores) gives 41 quick, which is every quick-tier step, and 45 full, or 47 with network access, which is every step the harness itself executes. The quick tier is CPU only; the full-tier figures include 4 GPU steps, the largest 16 GB VRAM, on public model weights. Those are planner figures, not runs: 37 of the 41 quick-tier steps carry a wall time measured on our machine on 2026-10-20 and those sum to about 10 min, and the tier was not re-timed at this commit. The measured outsider figure is separate: a clean-room replication executed 14 steps in about 6 min with nothing supplied beyond the repository, the organizers' data and the public references. ESM-1v and ESM-2 scoring took about 3 min, Evo 2 7B scoring of 198 calibration variants about 4 min and Borzoi about 7 min, each on one GPU. The whole pipeline takes about 6 h serial (about 4 h with the frequency lookup running alongside alignment) and 1.7 GPU-hours. The ranking alone (VCF in, ranked list out) needs no GPU and under 2 min of local compute plus the frequency lookup. There was no compute bill. Electricity for the pipeline was about 0.8 kWh (about 25 JPY, USD 0.17, at 31 JPY/kWh; GPU power sampled at about 200 to 270 W mean per job, 250 W assumed where not sampled); including exploratory and cross-check runs, about 4.1 GPU-hours and 2.3 kWh (about USD 0.47). Hardware cost is not amortised. On a rented GPU at USD 2.19/h the full pipeline would cost about USD 8 to 12. AI assistance ran on a flat-rate Claude Max subscription: about 35 h of active agent time over 14 days and about 2.0 M output tokens, with no per-token charge.

Provide a method abstract (up to 500 words). Please include strengths and limitations of your method as well as methodological detail.

Row 1 of our 10-row submission for proband `PROBAND01` is BUB1B c.2210T>G p.(Leu737Ter) with c.3006T>G p.(Asn1002Lys) (GRCh38 chr15:40209701 T>G and chr15:40220612 T>G), submitted with moderate confidence. Candidates were ranked by four rule versions, each hashed (sha256) before it ran and logged in a hypothesis register of 101 append-only entries; BUB1B is rank 1 under rules v3 and v4. The frozen rules

were then tested out of sample by writing public ClinVar Pathogenic/Likely pathogenic alleles into the child's own background: 1,611 ranking trials across eight pre-registered strata, plus a second pre-registered spike-in of 779 trials that chose the row-building rule (mean rank points 80.2 against 49.1 for our draft rule). What the ranking achieves is conditional, and the condition is measurable. In recessive genes outside the primary set that match at least one of the child's phenotype terms, rank-1 recall is 0.857 [0.814, 0.893] over 300 trials; in genes matching none it is 0.010 [0.003, 0.026] over 300. The same split holds inside the primary stratum, as a descriptive stratification that was not the pre-registered test: 160 of 163 trials (0.982) in phenotype-matched genes inside the ranking universe, and 0 of 121 in genes matching no term, every one of which landed at rank 2. The pre-registered test was the mixture of those two populations and its bar was not met: rank-1 recall 0.533 [0.477, 0.589] across the 14 of 18 MVA and phenocopy genes that held an eligible allele, against a pre-registered bar of 0.80, with a gene-clustered interval of [0.27, 0.80] and top-10 recall 0.947. Rank-1 recall for other BUB1B pairs is 1.000 over 100 trials. Four limits qualify all of this. Rank 1 depends on the ClinVar evidence tier: with ClinVar hidden no novel causal reaches rank 1 (0.000), although 0.790 reach the top 10. One MVA gene (RNU4ATAC) carries the highest phenotype score in the set and is never ranked at all, because it is non-coding and invisible to an exonic pipeline. Phase is not measured from short reads, so the compound-heterozygous configuration is a hypothesis. And the three calibrated variant-effect models disagree on p.Asn1002Lys. The EPCR values are calibrated from the spike-in.

Strengths: the four ranking rules (v1 to v4), the calibration rule, the two spike-in pre-registrations and the row-building pre-registration (ROW-MAP) were each hashed before they ran, the frozen rules were tested out of sample, and negative results are reported, including the missed robustness bar of 0.80. The structural-variant rule is the exception: its file was hashed after the unfiltered output of the same analysis was already visible in the session, so we cannot claim it was fixed before those data were seen. The outcome that rule governs is reported as NOT CONCLUDED in section 4. Limitations, in addition to the four stated above (the ClinVar dependence at rank-1 recall 0.000 among them): the consequence classifier never scores stop-loss, and five MVA genes cannot reach rank 1 because HPO does not annotate them with any of the child's terms.

Appendix. Register of tests run

Every test, check and attempt this project ran for Track 1, including the ones that returned nothing, were never run, or were superseded. It exists because a test that is left out of a report reads as a test that was never done. Items already discussed in the body carry a pointer to the section; items that appear only here carry their result in the line. Verdicts are the latest recorded for that question: where an early verdict was superseded by a later run, the later one is given.

The last column says what confirming that item needs. "Needs the alignment" means our BAM, which is not redistributable; "needs the child's call set" means the organizers' VCF, which a replicator has; "needs a GPU", "needs the network" and "needs a container" are the harness's own tiers.

#	Test	What was run	Verdict	Where, and what confirming it needs
1	Sex of the proband	Mapped-count ratios from the BAM index	Male; read counts consistent	Used by rules v2 to v4 (section 4). Needs the alignment index
2	CEP57 c.915_925dup11	Soft-clip and insertion counts against a background interval	RULED OUT at alignment level; blind to a mismatch-aligned duplication	Section 4. Needs the alignment
3	BUB1B upstream allele `rs576524605`	Read-level base count at the site	RULED OUT; reference at 45 reads, and the VCF has no record there	Section 4. Needs the alignment; the VCF half needs only the released call set

#	Test	What was run	Verdict	Where, and what confirming it needs
4	BUB1B intronic mobile element	Split, soft-clip and discordant rates against a 61 kb background, then structural calls in the locus	RULED OUT at both levels; not a cluster-level insertion caller, so a single weak insertion would be missed	Section 4. Needs the alignment and a structural caller
5	Dosage-sensitive structural variants, genome-wide	Manta 1.6.0 on the full alignment under a frozen rule, against the ClinGen dosage list	**NOT CONCLUDED.** 113 dosage-sensitive genes carry a passing call and none was carried to validation or phenotype matching	Section 4. Needs the alignment, a container and the ClinGen release, which is replaced at each release and has no archive URL for the one we used
6	Low-VAF mosaic signal	Per-chromosome fractions of low-VAF heterozygous calls	PARTIAL, then RULED OUT as copy-driven; the enrichment itself is unexplained and is most likely a mapping or paralog effect	Section 4. Needs the released call set only
7	Autozygosity	Longest het-free gap proxy	RULED OUT. A proxy, not a runs-of-homozygosity caller	Section 4. Needs the released call set only
8	ACMG secondary findings	ClinVar pathogenicity lookup only; the secondary-findings gene list was never applied	PARTIAL. Three Pathogenic/Likely pathogenic hits; the full screen was not run	Sections 8 and 11. Needs ClinVar
9	Arm-level depth	1 Mb binned depth, 12 bins per chromosome	RULED OUT at that resolution. Small changes are not excluded	Section 4. Needs the alignment
10	TRIP13 biallelic loss	Exon-level read of the canonical transcript	RULED OUT; no coding call. **Consequence: no positive control exists in this call set**	Sections 4 and 8. Needs the released call set only
11	CEP57 second allele	Canonical transcript records	RULED OUT for coding	Section 4. Needs the released call set only
12	Second-tier MVA genes	Exon-level reads plus 22 gnomAD v4 API lookups	RULED OUT except the BUB1B pair; only common or single calls	Sections 4 and 9. Needs the released call set and the gnomAD API
13	NAA10 hemizygous loss	Records plus depth in a male	RULED OUT	Section 4. Needs the alignment
14	SMN1 copy number	Two failed attempts, then marker positions from a full-gene comparison	Homozygous loss RULED OUT; **one copy against two NOT RESOLVED**, and would need an SMN-specific caller	Section 4. Needs the alignment and a specialised caller
15	Phase of the pair (P1)	Pre-registered read-backed phasing chain	**UNRESOLVED.** No template spans either gap; a code deviation from the pre-registered text is disclosed and changes nothing	Section 5. Needs the alignment; resolving it needs long reads or parental genotypes, which the dataset does not contain
16	De novo status	Not testable, no parental data	NOT TESTABLE	Sections 8 and 10. Needs parental genotypes
17	Function of the missense allele	Not testable, wet lab	NOT TESTABLE	Section 10. Needs a functional assay
18	M01, M02, M03 method hypotheses	**NOT RUN.** Each needs a positive control in the call set, and the TRIP13 control is absent (item 10)	NOT RUN, with the reason recorded rather than a negative result	This appendix. Running them needs a call set that contains a known causal, which this one does not

#	Test	What was run	Verdict	Where, and what confirming it needs
19	Genome-wide exonic frequency pass	87,198 calls classified against gnomAD v4.1 regional reads	Completed; three frequency classes	Section 4. Needs the released call set and the gnomAD files
20	Consequence step and its two recounts	Canonical-transcript consequence, recounted after the bug fix and again under replication	487 is the pre-fix first-transcript count and **does reproduce** under that definition; the corrected table gives 478 first-transcript or 481 any-transcript	Sections 4 and 8. Needs the released call set and GENCODE v44
21	Ranking v1	Frozen hashed rule, shuffled-label baseline, and a call-count ablation	Rank 8, shuffled baseline 7, and **not ranked at all under the unweighted call-count ablation**; three flaws recorded	Section 4. Needs the consequence table
22	Ranking v2	Frozen rule, same baselines	Rank 6, shuffled 4; data-only weighting does not separate BUB1B from background	Section 4. Needs the consequence table
23	Ranking v3	Frozen rule, ClinVar-shuffled baseline, v2-only ablation	Rank 1; rank 6 without the tier; leaves tier A when labels are shuffled. **Declared flaw, never fixed: tier A does not fit the inheritance mode of a recessive gene with one allele, and inheritance mode was not added to any later rule**	Section 4. Needs the consequence table and ClinVar
24	Ranking v4	Frozen rule plus dated Amendment 1, with phenotype-off, phenotype-only and shuffled ablations	Rank 1 in all of them. **The shuffled baseline is uninformative at this rule version, because tier A ordering dominates it**	Section 4. Needs the consequence table, ClinVar and the HPO release
25	v4 hash correction	Re-hash of the rule file after Amendment 1	The recorded hash covers the pre-amendment text; both hashes are recorded	Section 4. Needs the frozen rule file and its amended version, matched by their recorded hashes
26	Read-level check of the two row-1 calls	Depth, allele fraction, strand and mapping quality	Consistent with two real heterozygous calls. Says nothing about phase or pathogenicity	Section 1. Needs the alignment
27	Proof search	Dataset inventory, identifier resolution, literature pass, and a PubMed search on both allele names	Supported, not proven. **PubMed returns no record for either allele, so neither has a published clinical or functional report**; one submitted-variant identifier could not be resolved	Sections 1 and 10. Needs PubMed
28	External-service record	Log of every call that left the machine	Recorded. **The Ensembl release live at the coordinate lookups was not recorded**, so those are cited by access date	Section 9. Needs nothing
29	Input checksums and dataset commit	sha256 of the VCF, its index and the alignment; dataset commit confirmed in the browser	Byte-identical to the released commit; the later commit changed only the README	Section 9. Needs the released dataset

#	Test	What was run	Verdict	Where, and what confirming it needs
30	Readout detection limit	Depth coefficient of variation over 40 random 1 Mb bins per autosome	About 14% clonal fraction is the smallest whole-chromosome shift this data resolves. **Recorded consequence: an aggregate whole-genome aneuploidy-rate readout is below this detection limit and would need a different method**	Section 4. Needs the alignment
31	Copy-number rule v1	Frozen rule, 1 Mb bins, GC correction, clonal-fraction estimate	No chromosome called; the pre-registered falsifier was met	Sections 4 and 9. Needs the alignment and the frozen copy-number rule file, matched by its recorded sha256
32	Borzoi regulatory model	Local run against a 240-variant eQTL control set	AUC 0.496, chance level. **Not used**	Section 7. Needs a GPU and the public weights
33	Other regulatory models	**AlphaGenome is gated and could not be downloaded; Enformer was not run**	COULD NOT BE ATTEMPTED	This appendix. AlphaGenome needs access we do not have
34	Protein calibration, first pool	Run on a 1,626-row ClinVar control pool	**SUPERSEDED.** That pool does not follow the frozen calibration rule, so its AUCs are withdrawn and are not cited anywhere. The values in section 7 come from the rule-compliant pool built afterwards	This appendix and section 7. Needs ClinVar and a GPU
35	Protein calibration, compliant pool	98 missense controls under the frozen rule, bootstrap intervals	Three models informative, and the three disagree at the child's site	Section 7. Needs ClinVar, a GPU and the public weights
36	Evo 2 calibration	198 controls, FP8 disabled, one GPU, a pickle override, one transient crash; an early six-control smoke test was discarded	EXPLORATORY ONLY; no percentile reported. **Eligible counts differ from the rule text, recorded and not edited**	Section 7. Needs a GPU and the public weights
37	Consequence bug and its validation	Fix in a separate script, rerun of all four rules, validation on 6,000 public ClinVar SNVs	Rank 1 unchanged in all four; agreement with ClinVar's own terms rose from 0.589 to 0.994 on minus-strand transcripts	Section 8. Needs the released call set, GENCODE and ClinVar
38	Stop-loss blind spot	Code reading plus a 311-allele spike-in stratum	**CONFIRMED as a blind spot.** The classifier ignores the GENCODE stop_codon feature, so 0 of 311 were labelled stop_lost	Sections 6 and 8. The stratum's three counts are checked metrics of a harness step a reader can run. Needs GENCODE and ClinVar at the recorded release
39	FILTER column	Boolean check of the column on the submitted rows	Two rows use calls that fail FILTER; our candidate filter never read the column	Section 2. Needs the released call set only
40	Amendment 1 in the rerun script	Script diff plus an independent confirmation	The rerun script omits it; the submission applies the amendment as written and no other row moves	Section 4. Needs the rule file and the script

#	Test	What was run	Verdict	Where, and what confirming it needs
41	Spike-in of known causal events	1,611 ranking trials across eight pre-registered strata, pre-registered and hashed before any spiked ranking ran, then audited independently (83 of 83 rebuilt trials matched)	Bar not met at 0.533; ClinVar dependence total for rank 1; the control stratum separates; stop-loss invisible; EPCR vector adopted	Section 6. The strata, the EPCR vector and decisions D1 to D7 are checked metrics of harness steps a reader can run, and the trial tables are compared by sha256. Needs the consequence table and ClinVar at the recorded release, which the steps require before they will run
42	Spike-in stratum E	300 trials in recessive genes outside the MVA set that match at least one of the child's terms	**0.857 [0.814, 0.893] rank-1 recall**, against 0.010 for genes matching none. Rank 1 goes to a phenotype-matched tier-A gene, whatever the gene is	Section 6, table and the paragraph after the findings. Same needs as item 41
43	Row-building rule test	18 rule-by-construction combinations over 779 primary trials in 201 genes, with nine secondary strata, scored with the organizers' unmodified scorer	P5 with construction C3 adopted, +31.1 rank points; a near-tie among the replacements; row 1 identical under all 18	Section 6. Needs the row-map code and its aggregate results, together with the organizers' unmodified scorer
44	Scorer behaviour	The Space source read at a pinned commit, plus synthetic cases on invented coordinates	Order-only scoring; finding_type parsed but not used; the FAQ does not match the code	Sections 2 and 5. Needs the Space source, which is public
45	Replication harness, quick tier	A full quick-tier run at a pinned commit, 322 expected metric values	312 of 322 matched; six steps differed and produced errata E1 to E6; twelve steps did not run and are unverified	Section 9. Needs the harness and its own expected values; the twelve unrun steps need a GPU, the network or a container
46	Clean-room replication	An independent copy driven from the README alone, with a literal walkthrough, the quick tier and the test suite	**A literal README replication does not work.** An outsider runs 14 steps; every metric of every overlapping step was identical to our own run; five README defects named	Section 9. Needs the repository, the released dataset and the public references
47	Compute and cost audit	Read-only audit with every figure marked measured or estimated	Reported in full, including the figures that are estimates	Sections 9 and 11. Needs the tool logs
48	Citation work, four passes	A working ledger check, a record-level check of the cited identifiers, two independent audits of this report, and an adversarial identifier check	One ledger entry could not be found in any database and is cited nowhere; a claim that two second-tier genes had no supporting paper was checked, found to be wrong and removed; "biallelic loss" corrected throughout; one mechanism source replaced by a checkable one	References. Needs PubMed and Crossref

#	Test	What was run	Verdict	Where, and what confirming it needs
49	Adversarial review of this draft	A red-team pass over every cited claim in the draft, 13 attacks, plus a second-provider review at maximum effort	One blocking disclosure contradiction, six major and six minor items; the contradiction between "analysis stayed local" and the gnomAD lookup is fixed in section 9, and the objection to blanket ruled-out labels is why those tests now read as ruled out at the level tested	Section 9 and this appendix. Needs nothing
50	Outside critiques	Two de-identified questions put to outside models	Both said the proposal was not fundable as framed and that the central claim had to be narrowed. **One model returned figures that are not in the source it attributed them to; those figures were discarded and appear nowhere.** One curation page could not be retrieved and is unused	This appendix. Needs nothing; the discarded figures are recorded so the discard can be checked
51	Our own harness, checked by extending it	Building the new spike-in steps exercised an artifact that an existing step already compared with an exact-match rule	**DEFECT IN OUR OWN INSTRUMENT, found and fixed.** A single-file artifact was being resolved as a folder, so the exact-match comparison on it could never succeed, and `BLIND-01B`, the one existing step that relied on it, had been reporting its metrics NOT COMPARABLE when they were in fact identical. Fixed, with no expected value changed. **Reach, as verified:** that one step is the only pre-existing step whose exact-match rule names that artifact. Whether any earlier run's reported status was affected beyond it is **not established** here	This appendix. Needs the harness and a rerun of `BLIND-01B`

Acknowledgements

The challenge rules require the following acknowledgement in any publication or public communication arising from participation, and it is reproduced verbatim:

> "This work was made possible through the Hackathon, organized by Sage Bionetworks in partnership with the MVA Society, Hugging Face, and BEACON (The Benchmarking, Evaluation, and Assessment Consortium for Science), with prize sponsorship from AWS and Anthropic. We are deeply grateful to the child and their family who generously contributed their data and their story to advance research into this rare disease. We acknowledge their trust in making this Hackathon possible."

Dataset citation. The rules require the dataset to be cited using the reference provided on the challenge's Synapse page at the time of publication. No such reference was retrievable from the material available to us, so the dataset is cited here by its repository identifier: SageBio/mva-hackathon-2026-data at commit `59e322d` (VCF sha256 `bec64c68d3ad10d05b8071d07cdf87d217013d66d3ba02e24b505035c9d42175`), accessed 2026-10-17. If the Synapse reference is published, it supersedes this line.

References

Bibliographic records were retrieved by PMID from PubMed (NCBI E-utilities) and by DOI from Crossref on 2026-10-20. All 22 entries resolved against those records, and first author, year, venue, volume and pagination match in 22 of 22. Nine entries are cited in the text as author and year; the other thirteen are cited by identifier at the claim each supports.

Abdel-Salam GMH, Hellmuth S, Gradhand E, et al. 2023. Biallelic MAD2L1BP (p31comet) mutation is associated with mosaic aneuploidy and juvenile granulosa cell tumors. *JCI Insight* 8(22). PMID 37796616. doi:10.1172/jci.insight.170079.

Brixi G, Durrant MG, Ku J, et al. 2026. Genome modelling and design across all domains of life with Evo 2. *Nature* 652(8112):1349-1361. PMID 41781614. doi:10.1038/s41586-026-10176-5.

Chen X, Schulz-Trieglaff O, Shaw R, et al. 2016. Manta: rapid detection of structural variants and indels for germline and cancer sequencing applications. *Bioinformatics* 32(8):1220-2. PMID 26647377. doi:10.1093/bioinformatics/btv710.

Cheng J, Novati G, Pan J, et al. 2023. Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science* 381(6664):eadg7492. PMID 37733863. doi:10.1126/science.adg7492.

de Wolf B, Oghabian A, Akinyi MV, et al. 2021. Chromosomal instability by mutations in the novel minor spliceosome component CENATAC. *The EMBO Journal* 40(14):e106536. PMID 34009673. doi:10.15252/embj.2020106536.

Fujita H, Sasaki T, Miyamoto T, et al. 2020. Premature aging syndrome showing random chromosome number instabilities with CDC20 mutation. *Aging Cell* 19(11):e13251. PMID 33094908. doi:10.1111/acel.13251.

Grange LJ, Reynolds JJ, Ullah F, et al. 2022. Pathogenic variants in SLF2 and SMC5 cause segmented chromosomes and mosaic variegated hyperploidy. *Nature Communications* 13(1):6664. PMID 36333305. doi:10.1038/s41467-022-34349-8.

Guo J, He WB, Dai L, et al. 2024. Mosaic variegated aneuploidy syndrome with tetraploid, and predisposition to male infertility triggered by mutant CEP192. *HGG Advances* 5(1):100256. PMID 37981762. doi:10.1016/j.xhgg.2023.100256.

Hakonen AH, Karlberg S, Kiiski K, et al. 2026. Mosaic variegated aneuploidy as a novel feature in patients with Mulibrey nanism and TRIM37 variants. *Journal of Medical Genetics* 63(5):308-311. PMID 41702694. doi:10.1136/jmg-2025-111325.

Hanks S, Coleman K, Reid S, et al. 2004. Constitutional aneuploidy and cancer predisposition caused by biallelic mutations in BUB1B. *Nature Genetics* 36(11):1159-61. PMID 15475955. doi:10.1038/ng1449.

Lin Z, Akin H, Rao R, et al. 2023. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* 379(6637):1123-1130. PMID 36927031. doi:10.1126/science.ade2574.

Linder J, Srivastava D, Yuan H, et al. 2025. Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation. *Nature Genetics* 57(4):949-961. PMID 39779956. doi:10.1038/s41588-024-02053-6.

Meier J, Rao R, Verkuil R, et al. 2021. Language models enable zero-shot prediction of the effects of mutations on protein function. *bioRxiv (preprint)*. doi:10.1101/2021.07.09.450648.

Montfort M, Chabás A, Vilageliu L, et al. 2006. Analysis of nonsense-mediated mRNA decay in mutant alleles identified in Spanish Gaucher disease patients. *Blood Cells, Molecules & Diseases* 36(1):46-52. PMID 16326120. doi:10.1016/j.bcmd.2005.10.002.

Nagy E, Maquat LE. 1998. A rule for termination-codon position within intron-containing genes: when nonsense affects RNA abundance. *Trends in Biochemical Sciences* 23(6):198-9. PMID 9644970. doi:10.1016/s0968-0004(98)01208-0.

Ochiai H, Miyamoto T, Kanai A, et al. 2014. TALEN-mediated single-base-pair editing identification of an intergenic mutation upstream of BUB1B as causative of PCS (MVA) syndrome. *Proceedings of the National Academy of Sciences of the United States of America* 111(4):1461-6. PMID 24344301. doi:10.1073/pnas.1317008111.

Pinson L, Mannini L, Willems M, et al. 2014. CEP57 mutation in a girl with mosaic variegated aneuploidy syndrome. *American Journal of Medical Genetics. Part A* 164A(1):177-81. PMID 24259107. doi:10.1002/ajmg.a.36166.

Sieben CJ, Jeganathan KB, Nelson GG, et al. 2020. BubR1 allelic effects drive phenotypic heterogeneity in mosaic-variegated aneuploidy progeria syndrome. *The Journal of Clinical Investigation* 130(1):171-188. PMID 31738183. doi:10.1172/JCI126863.

Suijkerbuijk SJE, van Osch MHJ, Bos FL, et al. 2010. Molecular causes for BUBR1 dysfunction in the human cancer

predisposition syndrome mosaic variegated aneuploidy. *Cancer Research* 70(12):4891-900. PMID 20516114. doi:10.1158/0008-5472.CAN-09-4319.

Toledano I, Supek F, Lehner B. 2024. Genome-scale quantification and prediction of pathogenic stop codon readthrough by small molecules. *Nature Genetics* 56(9):1914-1924. PMID 39174735. doi:10.1038/s41588-024-01878-5.

Villarroya-Beltri C, Osorio A, Torres-Ruiz R, et al. 2022. Biallelic germline mutations in MAD1L1 induce a syndrome of aneuploidy with high tumor susceptibility. *Science Advances* 8(44):eabq5914. PMID 36322655. doi:10.1126/sciadv.abq5914.

Yang Y, Li N, Liu C, et al. 2025. Recessive SMC5 Variants in a Family with Near-Tetraploidy/Mosaic Variegated Aneuploidy. *Diagnostics* 15(23). PMID 41374403. doi:10.3390/diagnostics15233022.